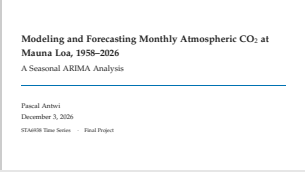
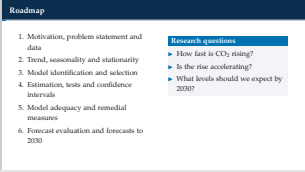
Modeling and Forecasting Monthly Atmospheric CO_2 at Mauna Loa, 1958–2026

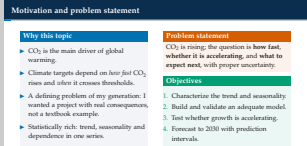
2026-11-01



Good morning. My project models and forecasts the monthly concentration of carbon dioxide measured at Mauna Loa from 1958 to 2026, and ends with a forecast to 2030.



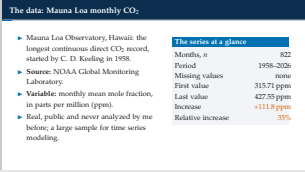
The talk follows the Box–Jenkins workflow: plot and transform, make the series stationary, identify and estimate a model, check it, and forecast. Three questions guide the analysis.



Why did I choose this topic? Carbon dioxide is the main driver of global warming, and climate targets depend on how fast it rises and when it crosses thresholds such as 440 ppm. Climate change is one of the defining problems of my generation, so I wanted my project to answer a question with real consequences rather than use a textbook example. The problem is not whether CO2 is rising; it is how fast, whether it is accelerating, and what to expect next, all with proper statistical uncertainty. My four objectives follow from that.

Modeling and Forecasting Monthly Atmospheric CO_2 at Mauna Loa, 1958–2026

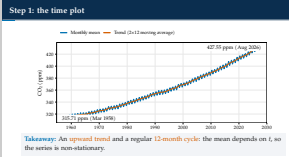
The data: Mauna Loa monthly CO_2



The data come from NOAA's Global Monitoring Laboratory. There are 822 consecutive monthly values from March 1958 to August 2026, with no gaps. Concentration rose from about 316 to 428 ppm, an increase of 35 percent.

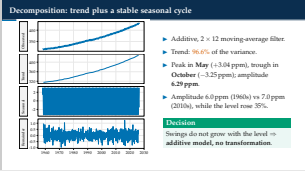
Modeling and Forecasting Monthly Atmospheric CO_2 at Mauna Loa, 1958–2026

Step 1: the time plot



The first step is always the time plot. Two features are obvious: a persistent upward trend and a seasonal cycle that repeats every year with almost the same shape. So the series is clearly non-stationary.

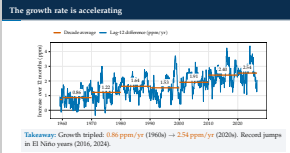
Decomposition: trend plus a stable seasonal cycle



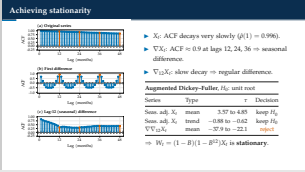
Classical decomposition separates the trend from the seasonal cycle. The cycle peaks in May and bottoms in October, driven by Northern Hemisphere vegetation. Because the amplitude barely changes while the level rises 35 percent, an additive model is adequate and no log transformation is needed.

Modeling and Forecasting Monthly Atmospheric CO_2 at Mauna Loa, 1958–2026

The growth rate is accelerating

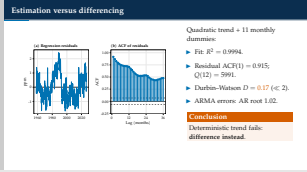


The lag-12 difference is the change over twelve months. Its decade averages tripled, so CO2 is not only rising but rising faster. The largest jumps occurred during El Nino events.

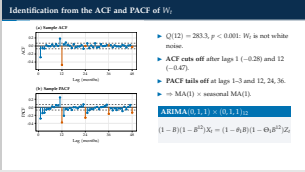


The ACF of the original series decays extremely slowly. After a first difference the seasonal lags stay near 0.9, so we also need a seasonal difference. The ADF test cannot reject a unit root for the seasonally adjusted series, but strongly rejects it after both differences. So the doubly differenced series is stationary.

Estimation versus differencing



As an alternative, I estimated the trend and seasonality by regression. The fit looks excellent, but the residuals wander for years: the Durbin–Watson statistic is 0.17, far below 2. Even with ARMA errors the AR root is essentially one, so differencing is the right approach.



The differenced series is not white noise, so we model it. The ACF cuts off after lag 1 and lag 12, while the PACF tails off at both the regular and seasonal lags. That is the signature of moving-average terms at both levels, which gives the multiplicative seasonal ARIMA zero-one-one by zero-one-one, period twelve.

Model selection: AIC and SBC

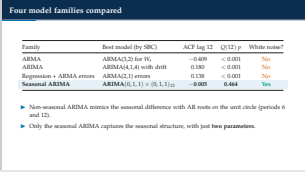
Model	Parameters	AIC	SBC	ΔSBC	Ljung–Box $Q(24)$ p
ARIMA$(0,1,1)\times(0,1,1)_{12}$	**2**	451.30	**465.39**	**0.00**	0.532
ARIMA$(1,1,1)\times(0,1,1)_{12}$	3	**447.58**	466.37	0.98	0.769
ARIMA$(0,1,2)\times(0,1,1)_{12}$	3	448.59	467.38	1.99	0.723
ARIMA$(0,1,1)\times(1,1,1)_{12}$	3	453.30	472.08	6.69	0.472
ARIMA$(2,1,0)\times(0,1,1)_{12}$	3	460.64	479.42	14.03	0.213
ARIMA$(1,1,0)\times(0,1,1)_{12}$	2	472.83	486.92	21.53	0.038
ARIMA$(0,1,1)\times(1,1,0)_{12}$	2	669.34	683.43	218.04	< 0.001

- Nine candidates fitted by maximum likelihood; **SBC** preferred in this large sample.
- Pure AR and seasonal-AR alternatives fit clearly worse.

I compared nine candidates, including AR, MA and seasonal AR terms. The two-parameter model has the lowest SBC. One richer model has a slightly lower AIC, but SBC, which penalizes extra parameters more in large samples, prefers the simpler model. Pure AR and seasonal AR versions are clearly worse.

Modeling and Forecasting Monthly Atmospheric CO_2 at Mauna Loa, 1958–2026

Four model families compared

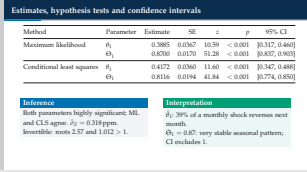


Four model families compared

Family	Best model (by SBC)	ACF lag 12	$Q(12)$ p	White noise?
ARMA	ARMA(3,2) for ∇t_t	−0.409	< 0.001	No
ARIMA	ARIMA(4,1,4) with drift	0.180	< 0.001	No
Regression + ARMA errors	ARMA(2,1) errors	0.138	< 0.001	No
Seasonal ARIMA	**ARIMA**$(0,1,1)\times(0,1,1)_{12}$	**−0.005**	**0.464**	Yes

- Non-seasonal ARIMA mimics the seasonal difference with AR roots on the unit circle (periods 6 and 12).
- Only the seasonal ARIMA captures the seasonal structure, with just **two parameters**.

I also fitted the best model from each family: ARMA, ARIMA, regression with ARMA errors, and seasonal ARIMA. Only the seasonal ARIMA leaves white-noise residuals. The others all leave strong correlation at lag 12. The non-seasonal ARIMA even puts its AR roots exactly on the unit circle to fake a seasonal difference.



Estimates, hypothesis tests and confidence intervals

Method	Parameter	Estimate	SE	z	p	95% CI
Maximum likelihood	θ_1	0.3885	0.0367	10.59	< 0.001	[0.317, 0.460]
	Θ_1	0.8700	0.0170	51.28	< 0.001	[0.837, 0.903]
Conditional least squares	θ_1	0.4172	0.0360	11.60	< 0.001	[0.347, 0.488]
	Θ_1	0.8116	0.0194	41.84	< 0.001	[0.774, 0.850]

Inference

Both parameters highly significant; ML and CLS agree. $\hat{\sigma}_\varepsilon^2 = 0.318$ ppm.
Invertible: roots 2.57 and $1.012 > 1$.

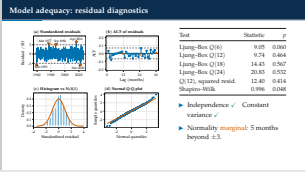
Interpretation

$\hat{\theta}_1$: 39% of a monthly shock reverses next month.
$\hat{\Theta}_1 = 0.87$: very stable seasonal pattern; CI excludes 1.

Both coefficients are highly significant, with z-statistics of 10.6 and 51.3, and the two estimation methods agree. All roots lie outside the unit circle, so the model is invertible. The seasonal coefficient of 0.87 means the seasonal pattern is very stable; because its confidence interval excludes one, seasonal differencing was justified.

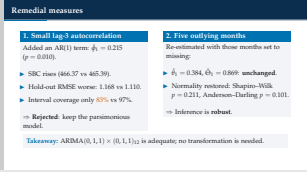
Modeling and Forecasting Monthly Atmospheric CO_2 at Mauna Loa, 1958–2026

Model adequacy: residual diagnostics



The residuals pass the Ljung–Box test at lags 6, 12, 18 and 24, and the squared residuals show constant variance. Normality is only marginally rejected, and that is driven by five unusual months, such as April 2016 during a strong El Nino.

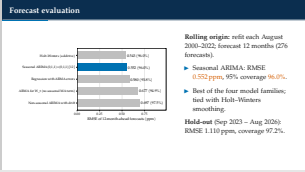
Remedial measures



I tried two remedies. Adding an AR term removed a small lag-3 correlation, but it raised the SBC and forecast worse, so I kept the simpler model. Setting the five outlying months to missing left the estimates unchanged and restored normality, so the inference is robust.

Modeling and Forecasting Monthly Atmospheric CO_2 at Mauna Loa, 1958–2026

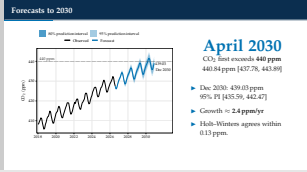
Forecast evaluation



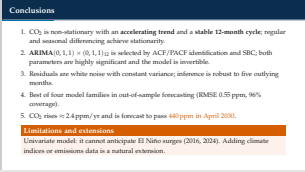
To test forecasting, I refit every model at 23 origins and forecast a year ahead each time. The seasonal ARIMA has an RMSE of about half a ppm with 96 percent interval coverage, close to the nominal 95. It is the best of the model families and essentially tied with Holt–Winters smoothing.

Modeling and Forecasting Monthly Atmospheric CO_2 at Mauna Loa, 1958–2026

Forecasts to 2030



The final model, fitted to all 822 months, forecasts that monthly CO2 will first exceed 440 ppm in April 2030. Growth continues at about 2.4 ppm per year, and an independent Holt–Winters forecast agrees to within 0.13 ppm.



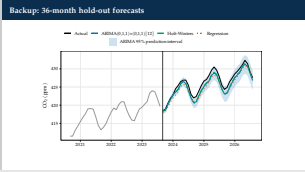
To conclude: the series needed both a regular and a seasonal difference; a simple two-parameter seasonal ARIMA fits it well, passes the diagnostics and forecasts best among the model families. CO2 is rising about 2.4 ppm per year and should pass 440 ppm in spring 2030. The main limitation is that a univariate model cannot anticipate El Nino surges.

Thank you
Questions?

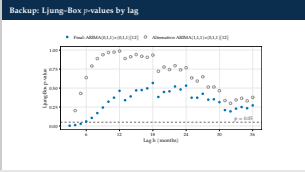
Thank you. I am happy to take questions.



Backup slide: the residual ACF of the best model in each family. Only the seasonal ARIMA, panel d, has no spikes at the seasonal lags.



Backup slide: forecasts made in August 2023 for the next three years. All methods under-predicted 2024, the year of record growth during the El Nino.



Backup slide: Ljung–Box p-values for lags up to 36 for the final and the alternative model.